

Data Analytics (DA) — Module 1: Excel Assignment 1 – Data Exploration | Formula

SCREENSHOTS TASK 1 — BASIC DATA EXPLORATION: SUM, COUNT, AVERAGE FIG 1: CELL M3 SELECTED — FORMULA BAR SHOWS =SUM(D2:D35)

The screenshot shows an Excel spreadsheet with a product data table and summary tasks. The formula bar at the top displays `=SUM(D2:D35)` for cell M3.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

Task 1 – Sum, Count & Average	
Total Price of All Products	10100
Count of Products	34
Average Price	297.0588235

Task 2 – Minimum & Maximum	
Minimum Price	30
Maximum Price	1000

Task 4 – SUMIF & COUNTIF	
Total Price – Electronics Category	8050
Products with Price < \$100	11

FIG 1: CELL M3 SELECTED — FORMULA BAR SHOWS =SUM(D2:D35) | RESULT: 10100

The screenshot shows the same Excel spreadsheet as Figure 1, but with cell M4 selected. The formula bar at the top displays `=COUNT(D2:D35)`.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
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05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
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FIG 2: CELL M4 SELECTED — FORMULA BAR SHOWS =COUNT(D2:D35) | RESULT: 34

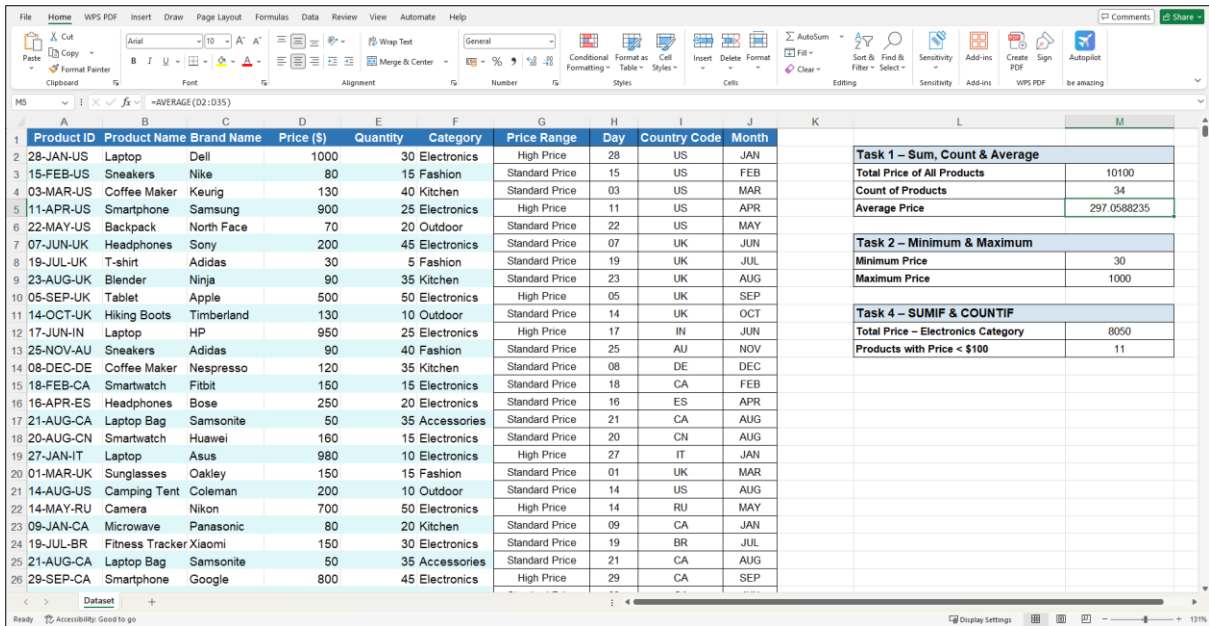


FIG 3: CELL M5 SELECTED — FORMULA BAR SHOWS =AVERAGE(D2:D35) | RESULT: 297.06

TASK 2 — MINIMUM & MAXIMUM VALUES: MIN, MAX

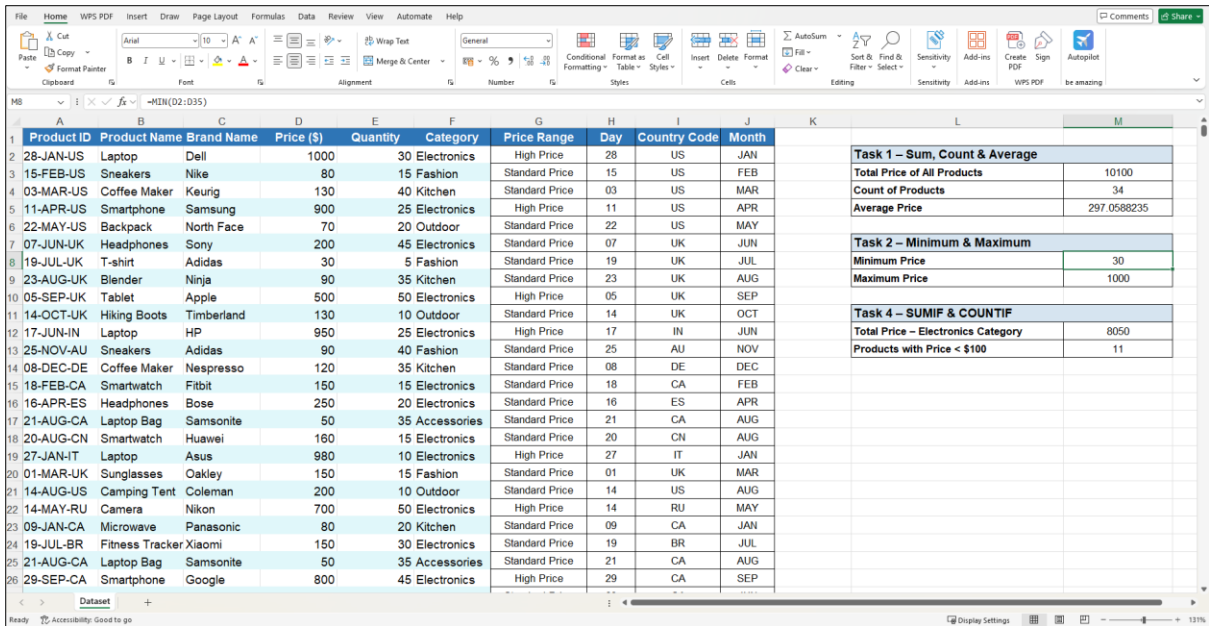


FIG 4: CELL M9 SELECTED — FORMULA BAR SHOWS =MIN(D2:D35) | RESULT: 30

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 5: CELL M10 SELECTED — FORMULA BAR SHOWS =MAX(D2:D35) | RESULT: 1000

Task 3 — Logical Function: IF (Price Range Column)

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
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14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
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14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 6: CELL G2 SELECTED — FORMULA BAR SHOWS =IF(D2>=500,"High Price","STANDARD PRICE") | APPLIED TO G2:G35

Task 4 — Conditional Functions: SUMIF & COUNTIF

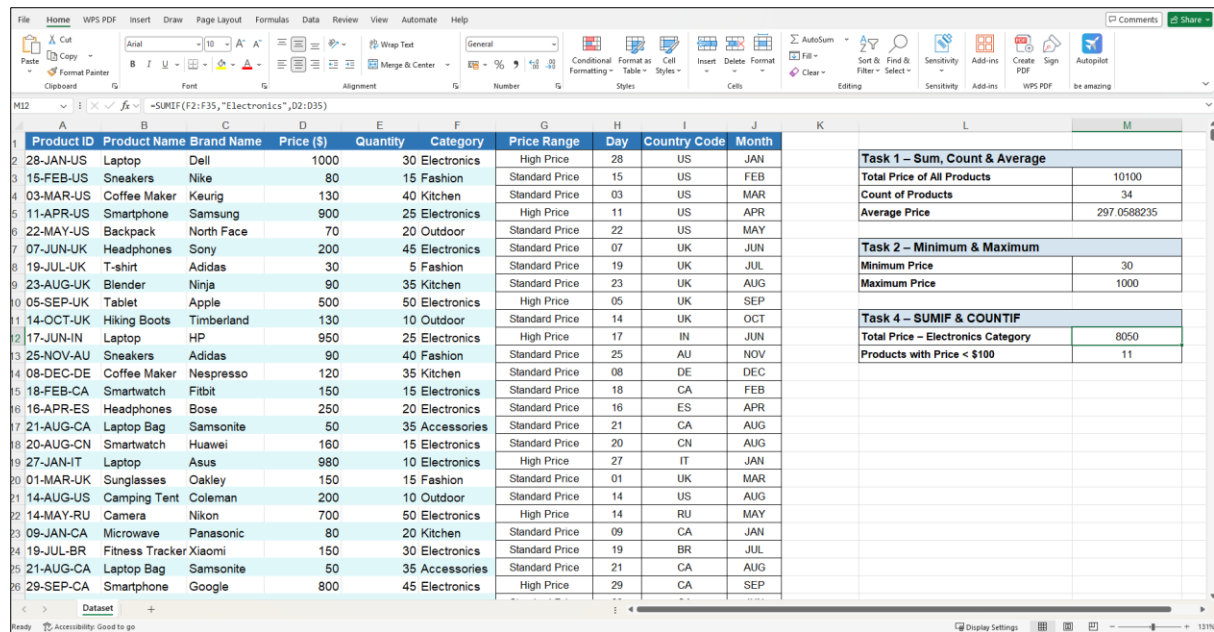


FIG 7: CELL M13 SELECTED — FORMULA BAR SHOWS `=SUMIF(F2:F35,\"Electronics\",D2:D35)` | RESULT: 8050

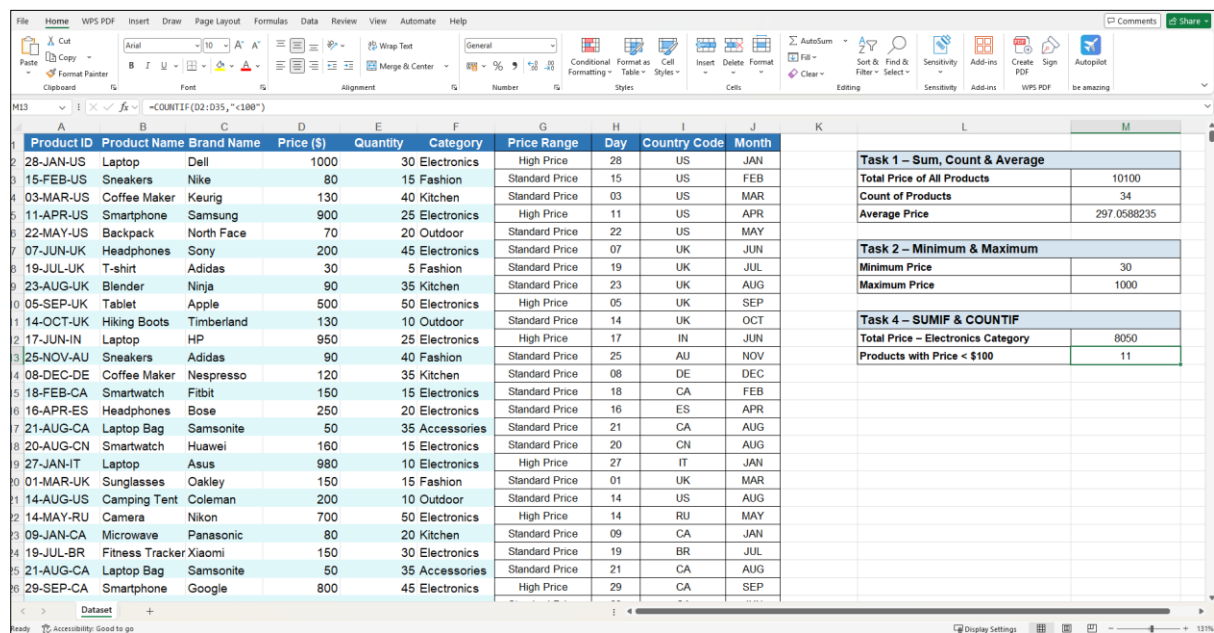
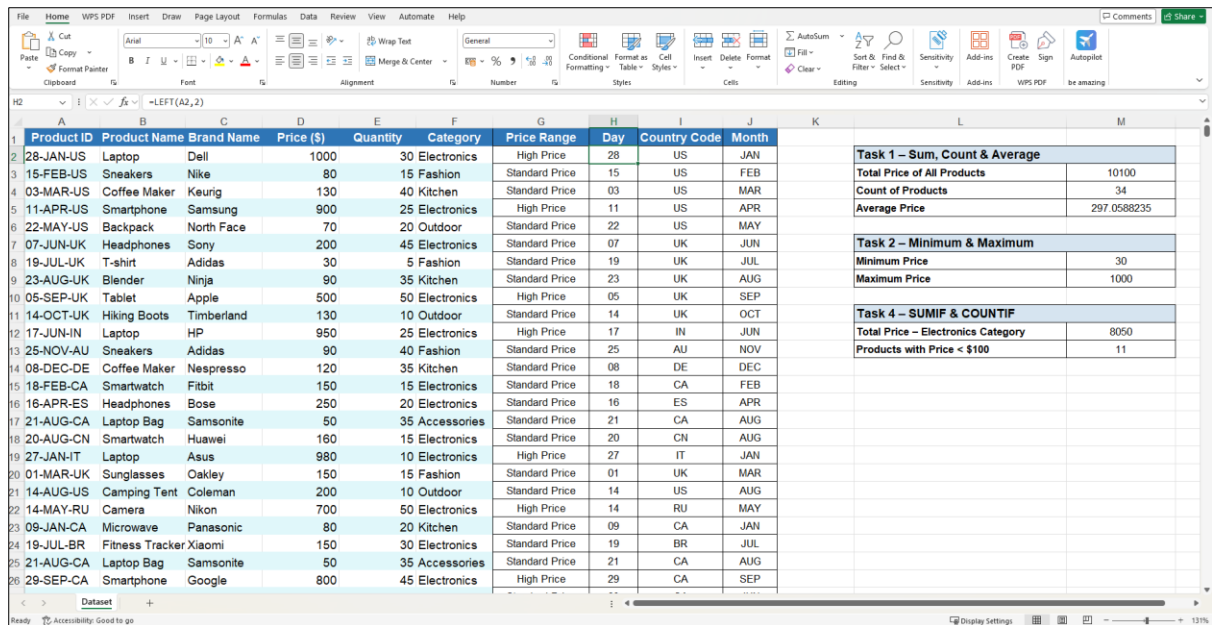


FIG 8: CELL M14 SELECTED — FORMULA BAR SHOWS `=COUNTIF(D2:D35,\"<100\")` | RESULT: 11

Task 5 — Text Functions: LEFT, RIGHT, MID



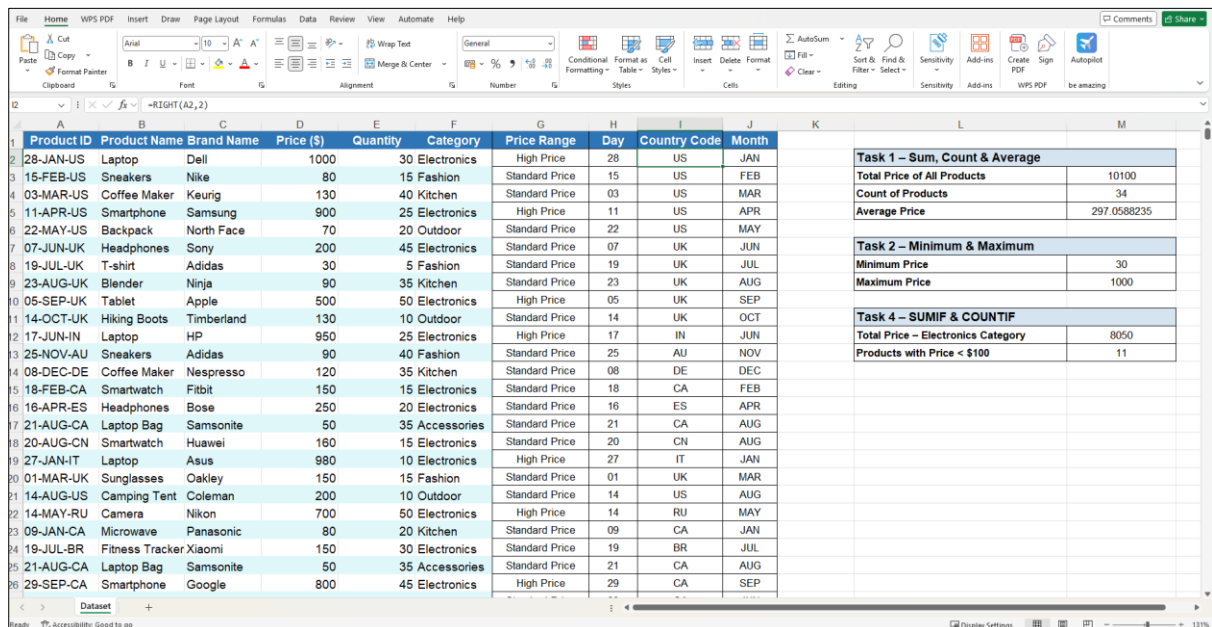
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14-MAY-RU	Camera	Nikon	700	60	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
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FIG 9: CELL H2 SELECTED — FORMULA BAR SHOWS =LEFT(A2,2) | EXTRACTS DAY (FIRST 2 CHARS) | APPLIED TO H2:H35



Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
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21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
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01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
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14-MAY-RU	Camera	Nikon	700	60	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
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FIG 10: CELL I2 SELECTED — FORMULA BAR SHOWS =RIGHT(A2,2) | EXTRACTS COUNTRY CODE (LAST 2 CHARS) | APPLIED TO I2:I35

